Punyakrit Singh Makhni

Full Stack Engineer | React, TypeScript, Node.js | AI and Agentic Systems
punyakritsinghmakhni@gmail.com
linkedin.com/in/punyakrit-singh-makhni | github.com/punyakrit | punyakrit.dev

SUMMARY

Full Stack Engineer with 3+ years of production software, 2 of them building for US startups. I own features end to end, from the requirement and design to the React and TypeScript interface, the Node.js APIs, the database and production. I've shipped LLM features and AI agents in real products: an AI marketing agent on live ad data, an MCP server that lets Claude and ChatGPT use my product, and a multi-model pipeline with validation gates.

TECHNICAL SKILLS

Frontend: React.js, Next.js, TypeScript, JavaScript (ES6+), HTML5, CSS3, Tailwind CSS, Responsive Design, Reusable Components, Performance Optimization
Backend & APIs: Node.js, Express.js, REST APIs, GraphQL, tRPC, WebSockets, Python (FastAPI), Authentication (OAuth 2.0, JWT), Third-Party Integrations
Databases: PostgreSQL (schema design, query optimization, full-text search), MongoDB, Redis, Supabase
AI & Agentic Systems: LLM APIs (OpenAI, Claude, Gemini), AI Agents, Tool Orchestration, Model Context Protocol (MCP), RAG, Embeddings, Structured Outputs, Prompt Orchestration, Evals
Quality & Delivery: Unit and Integration Testing, Code Review, Debugging, CI/CD, Docker, AWS (EC2, S3, CloudFront, Lambda), Cloudflare Workers, Git

PROFESSIONAL EXPERIENCE

RemJobs (remjobs.works) | *Software Engineer, own product* | Remote — Aug 2026 – Present

- Built and run, alone, a full-stack job search over **400K+ open roles from 250K+ startups**, with integrations into **20+ hiring platforms**. Launched Sep 2026; ~200 daily visitors and first paying users in Oct 2026.
- Shipped every feature end to end: search with filters, sign-in and paid plans (Next.js, TypeScript, Python, PostgreSQL).
- Built an **MCP server** so AI agents in Claude, ChatGPT and Cursor can call RemJobs as a tool, secured with OAuth 2.0 and per-plan limits checked on every call.
- Made the data pipeline fail safely (a crashed run writes nothing and the next run resumes from the last good snapshot), and served search from a warm in-memory index in **8–16 ms**.

Gradly | *Software Engineer* | Remote (United States) — Mar 2026 – Jul 2026

- Owned the signup and account flow **15,000+ international students** used to buy US health insurance: the React and Next.js UI in TypeScript and the Node.js REST APIs behind it, over personal and policy data.
- Built the internal tools and admin dashboards the operations team used every day, on GraphQL and FastAPI services over PostgreSQL and Redis.

TheFocus.AI | *Software Engineer (Contract)* | Remote — Dec 2025 – Mar 2026

- Built semantic search over member profiles with embeddings, from the retrieval API to the search UI.
- Cut page load times by ~**80%** for an invite-only professional network by reworking rendering, auth and infrastructure.

Muze AI | *Software Engineer* | Remote (United States) — Sep 2024 – Oct 2025
Software Engineer Intern, Jun 2024 – Aug 2024

- Built **Muze CMO** from scratch: an AI marketing agent that reads live Meta Ads and Google Ads data, predicts whether a creative will work before spend, and generates the next ad, grounded by a RAG layer on Supabase (PostgreSQL) and Redis.
- Lifted engagement **25%** with new React features, cut API latency by **40%** through query optimization and caching, and took releases from every ~3 days to **under 12 hours** with CI/CD and Docker on AWS.

PROJECTS

AI Demo Video Builder | *Python, FastAPI, Pydantic, Gemini* — github.com/punyakrit/Ai-demo-video

- Multi-model AI workflow (4 models, 8 stages): the LLM plans, plain code executes. Nothing renders until the plan passes **32 validation checks**; **410 automated tests**.

InfiniteUGC | *LLMs, video models, AWS* — infiniteugc.com

- Built end to end, including the UI: AI video ads for **50+ brands in 32+ languages**. Long-running jobs retry on their own when model providers fail.

EDUCATION

Lovely Professional University | Bachelor of Technology (B.Tech) in Computer Science and Engineering